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Supplementary Table 1: Search strategy as applied in Medline (date last run 16.06.2019)

1. exp Critical Illness/
2. exp Critical Care/
3. exp Intensive Care Units/
4. critical care.mp.
5. critical illness*.mp.
6. intensive care.mp.
7. ICU.mp.
8. ITU.mp.
9. OR/1-8
10. exp Patient Participation/
11. patient involve*.mp.
12. patient engage*.mp.
13. shared decision making.mp.
14. patient participat*.mp.
15. service user involve*.mp.
16. family involve*.mp.
17. family participat*.mp.
18. family engage*.mp.
19. caregiver involve*.mp.
20. caregiver engage*.mp.
21. caregiver participat*.mp.
22. relative involve*.mp.
23. relative engage*.mp.
24. relative participat*.mp.
25. OR/11-24
26. 9 AND 25

Xyrichis A, *et al. BMJ Open* 2021; 11:e042556. doi: 10.1136/bmjopen-2020-042556

Supplementary File: Outcomes grouped by type of family involvement intervention

Supplementary Table 2: Participation in medical rounds interventions

Study	Allen et al. 2017	Jacobowski et al. 2010	Weber et al. 2018
Intervention / Measure	Family on rounds	Family rounds	Family rounds
Communication (improvement, % agree)	Pre: n=49 (100%) Post: n=47 (100%) Change 0% p=0.68 RR: 1, CI: 0-0		
Family knowledge	Improvement, % agree Pre: n=146 (35%) | Post: n=374 (88%) Change 52% p<0.001 RR: 2.46, CI: 2.14-2.81		
Nurse workload	Reduction, % agree Pre: n=32 (65%) | Post: n=45 (96%) Change 31% RR: 1.46, CI: 1.18-1.81		
Physician satisfaction	Pre: n=4 (80%) | Post: n= 5 (83%) Change 3% RR: 1.04, CI: 0.59-1.83		
Frequency nurse communication		% highest score Pre: 64% | Post: 57% Change: -7% p=0.30 RR: 0.89 CI: 0.70-1.12	
Frequency doctor communication		% highest score Pre: 38% | Post: 60% Change: 22% p=0.004 RR: 1.60 CI 1.18-2.17	
Included in decision making		% highest score Pre: 66% | Post: 76% Change: 10% p=0.12 RR: 1.15 CI 0.96-1.38	
Support in decision making		% highest score Pre: 49% | Post: 69% Change: 20% p=0.005 RR: 1.39 CI 1.09-1.79	
Time to ask questions		% highest score Pre: 40% | Post: 23% Change: -17% p=0.02 RR: 0.57 CI 0.37-0.90	
FS ICU 24 - Frequency doctor communication			% top scores Control: 43% Intervention: 56% Change: 13% p>0.05 RR: 1.31 CI 0.90-1.89
FS ICU 24 - Frequency nurse communication			% top scores Control: 62% Intervention: 78% Change: 16% p>0.05 RR: 1.24 CI 0.98-1.58
FS-ICU (24) - Decision making			Mean score (SD) Control: 85.1 (16.3) Intervention: 88.6 (14.6) Change: 3.5 p=0.20 CI: -1.98-8.98
FS-ICU (24) - Global score			Mean score (SD) Control: 86.0 (16.0) Intervention: 90.8 (10.7) Change: 4.8 p=0.20 CI: -0.12 - 9.72
Support given (emotional)			% top scores Control: 54% Intervention: 71% Change: 17% p>0.05 RR: 1.32 CI 0.99-1.75
Coordination of care			% top scores Control: 68% Intervention: 77% Change: 9% RR: 1.13 CI 0.91-1.41

Supplementary Table 3: Multi-component family involvement interventions

Study	Cray 1989	Davidson et al. 2010	Marshall et al. 2016	White et al. 2018
Intervention / Measure	Family programme	Facilitated sensemaking	Nutrition intervention	Multi family support
Satisfaction with family involvement	100% agreed (76/76)	Observations showed family members were most engaged when receiving information about how to participate at the bedside.	Reports of increased family involvement, enabling patient advocacy.	
Helpful in understanding loved one's illness	93% agreed (71/76)		Improvement in knowledge regarding nutritional needs.	
HADS - global score (0-42)				Mean score Control: 12.0 Intervention: 11.7 Change: -0.34 p=0.61 CI −1.67 - 0.99
QOC - clinician-family communication (0-100)				Mean score Control: 62.7 Intervention: 69.1 Change: 6.39 p=0.001 CI 2.57 - 10.20
PPPC - Patient Perception of Patient Centeredness (1-4)				Mean score Control: 1.8 Intervention: 1.7 Change: -0.15 p=0.006 CI −0.26 - −0.04
Length of stay - Hospital				Mean score Control: 6.7 Intervention: 7.4 Change: 0.90 p=0.045 CI 0.81 - 1.00
Length of stay - ICU				Mean score Control: 13.5 Intervention: 10.4 Change: 0.77 p<0.001 CI 0.69 - 0.87
IES - PTSD (0-88)				Mean score Control: 20.3 Intervention: 21.2 Change: 0.90 p=0.49 CI −1.66 - 3.47

Supplementary Table 4: Discussion based family involvement interventions

Study	Almoosa et al. 2009	Hollman et al. 2018	Garrouste et al. 2016	Huang et al. 2018	Curtis et al. 2016	Shaw et al. 2014
Intervention / Measure	CPR discussions	Health conversations	Interprofessional family conference	Primary physician involvement	Communication facilitator	Team training
Satisfaction with CPR Discussion	Questions answered 84% (36/43) CI 70-91 Facilitated decision 77% (33/42) CI 64-88					
Time to decision	Days mean (sd) Control 7.4 (3.2) Intervention 6.6 (6.5) Change -0.8 days CI -1.48-3.08					
Death	Control 8% (3/39) Intervention 37% (17/45) Change 31% RR 4.91 CI 1.55-15.51					
Change to less aggressive preference	Control 18% (8/39) Intervention 25% (11/45) RR: 1.19 CI 0.53-2.66					
Inclusion in decision making		Qualitative findings of greater family engagement with ICU nurses.		% completely satisfied Control: 61.4% Intervention: 75.9% Change: 14.5% p=0.05 RR: 1.23 CI 1.03-1.49		

Study	Almoosa et al. 2009	Hollman et al. 2018	Garrouste et al. 2016	Huang et al. 2018	Curtis et al. 2016	Shaw et al. 2014
Intervention / Measure	CPR discussions	Health conversations	Interprofessional family conference	Primary physician involvement	Communication facilitator	Team training
HADS - anxiety			At 90 days, Median [IQR] Control: 8 [4.5-12] Intervention: 4 [1-9] Change: 4 p=0.01 % sign. Anxiety (>8) Control: 52.3% (n=23) Intervention: 33.3% (n=14) Change: 19% p=0.08 RR: 0.95, CI 0.63-1.44			
HADS - depression			At 90 days, Median [IQR] Control: 5.5 [1-11.5] Intervention: 2 [0-6] Change: -3.5 p=0.04 % sign. Depress. score > 8 Control: 38.6% (n=17) Intervention: 23.8% (n=10) Change: 14.8% p=0.14 RR: 0.61, CI 0.31-1.18			
PDEQ			Median [IQR] Control: 14.5 [11-23] Intervention: 13 [0-17] Change: -1.5 p=0.17			
IES-R			At 90 days, Median [IQR] Control: 24 [12.5-45] Intervention: 21 [9-33] Change: -3 p=0.24			
FS-ICU (24) - ICU Care				Mean (sd) Control: 88.38 (14.5) Intervention: 90.22 (12.68) Change: 1.84 p=0.28 CI -2.42 - 6.09		
FS-ICU (24) - Global score				Mean (sd) Control: 84.91 (12.17) Intervention: 86.4 (11.76) Change: 1.49 p=0.16 CI -2.14 - 5.12		
FS-ICU (24) - Decision making				Mean (sd) Control: 80.07 (12.76) Intervention: 81.06 (15.1) Change: 0.99 p=0.16 CI -3.00 - 4.98		
Control over patient care				% completely satisfied Control: 55.6% Intervention: 73.6% Change: 18% p=0.02 RR: 1.31 CI 1.07-1.61		
GAD 7 - anxiety					Mean score 3 months Control: 3.0 Intervention: 2.3 Change: -0.7 p=0.50 CI: -2.91 - 1.42 Mean score 6 months Control: 2.7 Intervention: 1.8	

Study	Almoosa et al. 2009	Hollman et al. 2018	Garrouste et al. 2016	Huang et al. 2018	Curtis et al. 2016	Shaw et al. 2014
Intervention / Measure	CPR discussions	Health conversations	Interprofessional family conference	Primary physician involvement	Communication facilitator	Team training
					Change: -0.9 p=0.43 CI: -3.10 - 1.32	
PHQ 9 - depression					Mean score 3 months Control: 4.9 Intervention: 3.1 Change: -1.8 p=0.09 CI: -3.89 - 0.31 Mean score 6 months Control: 4.7 Intervention: 2.4 Change: -2.3 p=0.01 CI: -4.30 - -0.42	
PCL - PTSD					Mean score 3 months Control: 31.6 Intervention: 29.8 Change: -1.7 p=0.47 CI: -6.65 - 3.12 Mean score 6 months Control: 30.6 Intervention: 27.1 Change: -3.5 p=0.056 CI: -7.12 - 0.09	
Staff confidence. All items improved, biggest change: confidence in encouraging family presence						Mean score (SD) Pre: 7.0 (1.7) Post: 8.4 (1.1) Change: 1.4 p<0.001 CI: 0.81-1.99
FS ICU 24 - Frequency nurse communication						Mean score Pre: 79.2 Post: 87.18 Change: 7.98 p=0.04
FS ICU 24 - Frequency doctor communication						Mean score Pre: 67.86 Post: 76.69 Change: 8.83 p=0.04
FS ICU 24 - Honesty of information						Mean score Pre: 77.78 Post: 87.08 Change: 9.30 p=0.01
FS-ICU (24) - Decision making						Mean score Pre: 77.47 Post: 83.32 Change: 5.85 p=0.05
FS-ICU (24) - Global score						Mean score Pre: 83.21 Post: 85.69 Change: 2.48 p=0.32

Supplementary Table 5: Web-based support interventions

Study	Dalal et al. 2015	Dykes et al. 2017	Ernecoff et al. 2016	Huffines et al. 2013
Intervention / Measure	Patient centred toolkit	Patient engagement ICT	Tablet decision support	Support care algorithm
Usability	Mean (sd) usability score m 74 (16.7)			
Satisfaction	Satisfaction 72% (13/18) Mean (sd) satisfaction score 4.06 (0.94)			

Study	Dalal et al. 2015	Dykes et al. 2017	Ernecoff et al. 2016	Huffines et al. 2013
Intervention / Measure	Patient centred toolkit	Patient engagement ICT	Tablet decision support	Support care algorithm
HCAHPS		% top score 9-10 Pre: 71.8%, n=53, CI 61.1-82.6 Post: 93.3%, n=58, CI 88.2-98.4 Change 21.5 RR 1.33 CI 1.10-1.55		
FS-ICU (20)		Mean score Pre: 84.3, n=106, CI 81.3-87.3 Post: 90, n=156, CI 88.1-91.9 Change: 5.7, CI 2.31-9.09		
Adverse events		Pre: 59/1000 patient days Post: 42/1000 patient days Change 17, CI 6.95-27.05 Relative reduction 29%		
Length of stay		Mean (median) LoS [range] Pre: 4.9 (2) [1-108], n=881 Post: 5.0 (2) [1-115, n=904 p=0.61		
Share in decisions about care planning			Qualitative perceptions of improving family engagement, communication and involvement in decision making.	Mean satisfaction (1-4) Pre: 2.97 | Post: 3.41 Change: 0.44 p=0.07 % scoring excellent Pre: 45% | Post: 68% Change: 23% p=0.009 RR: 1.52 CI 1.03-2.24
Support and encouragement given				Mean satisfaction (1-4) Pre: 3.38 | Post: 3.52 Change: 0.15 p=0.47 % scoring excellent Pre: 60% | Post 75% Change: 15% p=0.14 RR: 1.23 CI 0.91-1.65
Nurses and doctors working together as a team				Mean satisfaction (1-4) Pre: 3.38 | Post: 3.73 Change: 0.35 p=0.04 % scoring excellent Pre: 64% | Post: 83% Change: 19% p=0.04 RR: 1.31 CI 1.00-1.71

Supplementary Table 6: Environmental unit changes & Participation in care interventions

Study	Choi et al 2013	Rippin et al. 2015	Prichard & Newcomb 2015
	Environmental unit changes		Physical participation in care
Intervention / Measure	PFC Unit design	Family centred design	Hand massage
Family interaction with patient	r=0.632 | p<0.01		
Family interaction with staff	r=0.552 | p<0.01	Qualitative observations of increased, family-initiated interactions.	
HADS - anxiety			Mean change in score Control: -0.4 Intervention: -3.87 Change: -3.47 p=0.002 CI -5.5 - -1.4
HADS - depression			Mean change in score Control: -0.3 Intervention: -2.5 Change: -2.2 p=0.10 CI -0.49 - 4.7